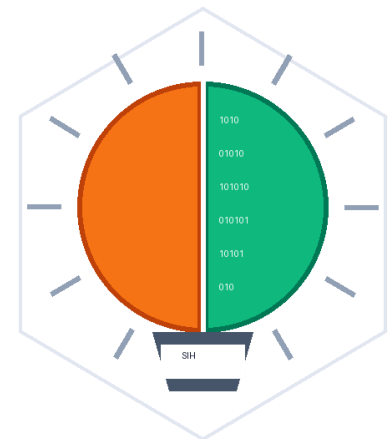




SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID – 26076
- Problem Statement Title- Development of personalized homepage for 'Mausam' mobile application.
- Theme- Smart Automation
- PS Category- Software
- Team ID- [Your Team ID]
- Team Name (Registered on portal) : Mausam 2.0





Detailed Explanation of Proposed Solution:

A mobile-based AI solution for automated and personalized weather intelligence.

Users access tailored meteorological insights through the Mausam 2.0 PWA application.

Smart automated intelligence and scoring models compute and classify:

- Health & AQI (Asthma Triggers, UV Sunburn)
- Outdoor Fitness (Best Running Hours, Heat Stress)
- Beach & Surfing (Tide Harmonics, Swell Height)
- Travel & Commute (AI Packing, Road Visibility)
- Agriculture (Topsoil Moisture %, Spray Windows)

Transforming static raw weather data into an adaptive, personalized, and actionable daily intelligence experience.

The system estimates health risks, workout scores, and crop water balance from real-time feeds. Results are processed through a Python 3 backend gateway and cached securely in local storage.

Generates an instant digital quality forecast report with personalized recommendations and voice briefings.

How It Addresses the Problem?

- Reduces subjectivity and confusion in raw weather data.
- Applies lifestyle-specific criteria across 8 citizen personas.
- Automatically identifies health hazards (asthma, heatstroke).
- Provides proactive commute, school transit, and packing alerts.
- Enables faster decisions compared with manual chart inspection.
- Generates digital and traceable forecasts, reducing weather uncertainty.

Innovation & Uniqueness of the Solution

- Smart Auto-Blend engine dynamically shifts homepage across the day.
- Automated detection of workout windows using WBGT heat models.
- AI-powered Smart Packing matrix integrated into travel workflow.
- Mobile-first PWA approach enables access directly on any device.
- Combines live weather + AI scoring + backend caching + voice TTS.
- Maintains offline forecast caching for seamless resilience.



Tech Stacks Used

Artificial Intelligence and Scoring:

- Open-Meteo & IMD Synoptic Data APIs
- CPCB National Air Quality Index (PM2.5, PM10, NO2, O3)
- INCOIS Marine Swell & Agromet Soil Moisture (0-7cm)

Mobile Application:

- HTML5, CSS3, Modern Tailwind CSS Framework
- Glassmorphism UI with 1-Click Mobile Simulator
- Progressive Web App (PWA) Offline Architecture

Backend and Database:

- Python 3 Standard Library (http.server, urllib, json)
- 5-Minute Memory Cache & Browser LocalStorage

Visualization & Voice:

- Chart.js Hardware-Accelerated Canvases
- W3C Web Speech API (English & Hindi TTS)

App Workflow





FEASIBILITY AND VIABILITY

Feasibility of Idea:

Technical Feasibility:

- Built using lightweight Python 3, Vanilla JS & Chart.js.
- Zero-dependency backend eliminates heavy database licensing.

Operational Feasibility:

- Simple workflow: Location → Fetch → Score → Dashboard.
- Automated algorithms reduce manual effort and processing time.
- Offline caching enables easy tracking during network drops.

Deployment & Scalability:

- Cloud-compatible micro-gateway for real-world deployment.
- Can scale to more users, concurrent requests, and datasets.

Field Applicability:

- Designed for quick access across smartphones and browsers.
- Requires minimal equipment beyond a standard smartphone.

Challenges, Risks & Mitigation:

API Rate Limits & Latency:

- External APIs may experience latency spikes during peak events.
- Mitigation: 5-minute memory caching and proxy deduplication.

Microclimate & Terrain Quality:

- Diverse Indian terrain may show localized microclimate variations.
- Mitigation: Pinpoint GPS geocoding and 36 States/UTs directory.

Connectivity & Deployment:

- Unstable connectivity in rural farm fields can affect live fetching.
- Mitigation: PWA offline caching ensures reliable local recall.

User Persona Overlap & Routine:

- Citizens may fit multiple roles (parent, commuter, runner).
- Mitigation: Smart Auto-Blend schedules modules by routine.

**Consistent & Personalized Forecasting:**

Impact: Transforms raw numbers into lifestyle-specific actionable advice.

Benefit: Reduces cognitive burden and empowers proactive citizen decisions.

Public Health Protection:

Impact: Correlates humidity with PM2.5/PM10 and monitors UV sunburn timers.

Benefit: Minimizes asthma attacks, respiratory distress, and heat exhaustion.

Agricultural Optimization:

Impact: Provides topsoil (0-7cm) moisture % and foliar spraying suitability windows.

Benefit: Prevents pesticide runoff and optimizes crop irrigation schedules.

Travel & Commute Safety:

Impact: Analyzes airport fog delay risks, road visibility distance, and waterlogging.

Benefit: Prevents traffic accidents, flight missed connections, and packing errors.

Digital & Audio Inclusion:

Impact: Delivers natural voice weather bulletins in English and हिन्दी (Hindi).

Benefit: Bridges the digital literacy divide for rural farmers and elderly citizens.

ACTIONABLE METEOROLOGY.**A SAFER TOMORROW.**

Better Health | Higher Farm Yield | Safe Commute

Public Health Shield

Minimizes asthma & heatstroke risks.

Higher Farm Yield

Optimizes irrigation & foliar spraying.

Transit & Safety

Fog visibility & waterlogging alerts.

Environmental Care

Less fertilizer waste & runoff.

Universal Audio Inclusion

Natural English & Hindi voice briefings increase citizen trust and reach.



DATA RESEARCH:

- <https://open-meteo.com/> (Synoptic Meteorological & Soil Moisture API)
- <https://cpcb.nic.in/> (Central Pollution Control Board - National AQI Data Pipeline)
- <https://incois.gov.in/> (Indian National Centre for Ocean Information Services - Coastal Tides & Swell)
- <https://mausam.imd.gov.in/> (India Meteorological Department - Agromet GKMS Bulletins)

TECHNICAL RESEARCH:

- https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API (W3C Web Speech API)
- <https://www.chartjs.org/> (Chart.js Hardware-Accelerated Data Visualization)
- <https://tailwindcss.com/> (Tailwind CSS Framework)
- <https://docs.python.org/3/library/http.server.html> (Python 3 HTTP Server)